大量科技
TALIANG TECHNOLOGY

FVI - SECS/GEM

User manual

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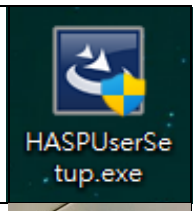
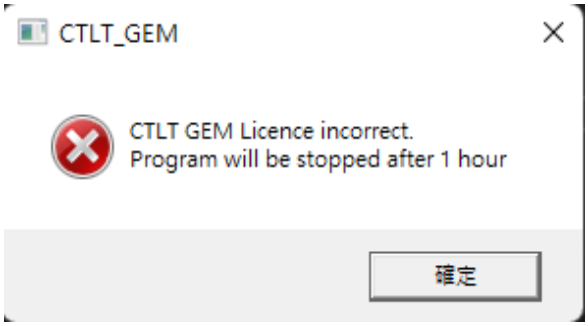
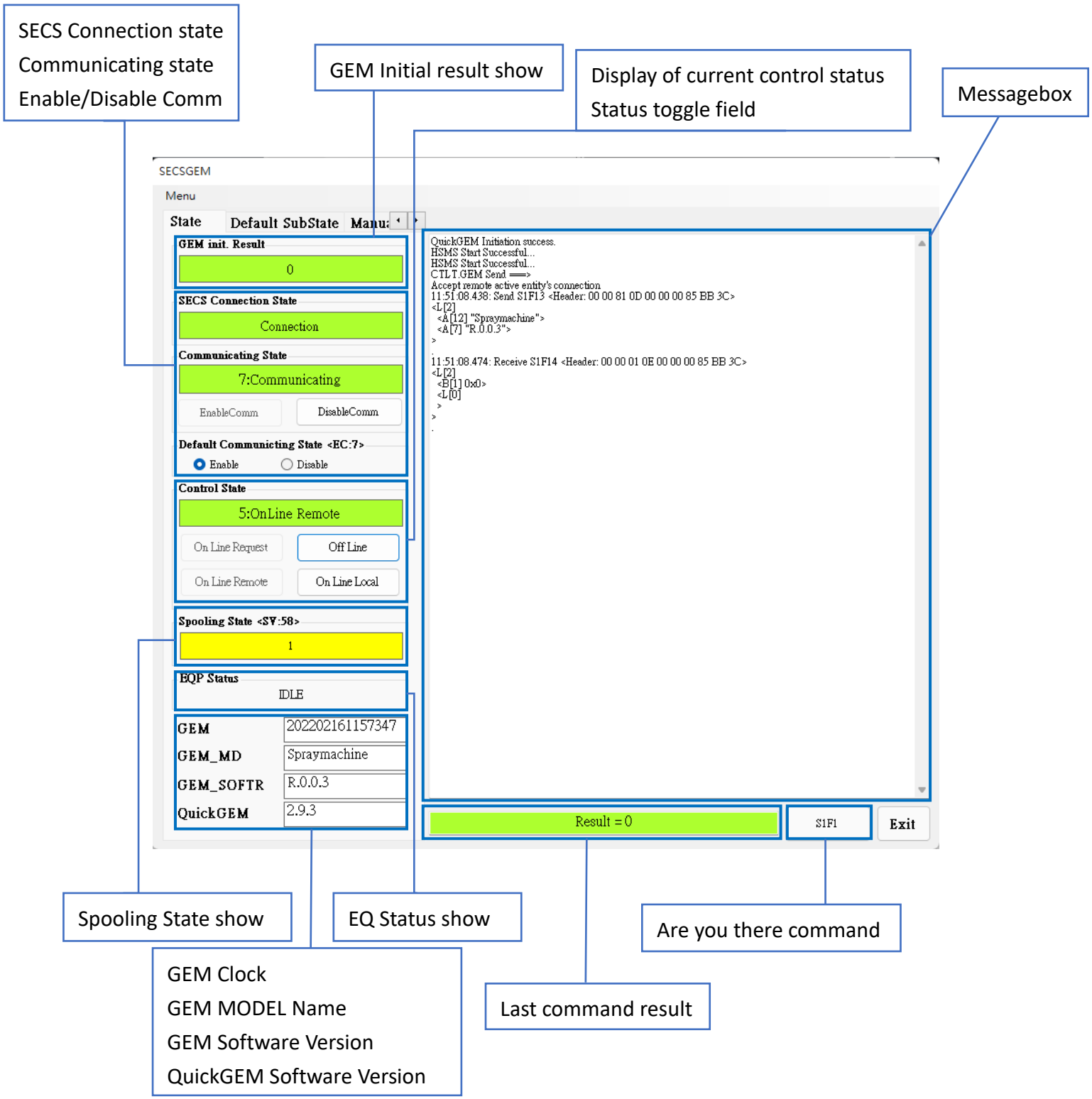
SECS/GEM Driver & Dongle Install

STEP1 : **Install HASPUserSetup.exe**	
STEP2 : **Insert SECS/GEM Dongle to Computer's USB port**	

※If the installation fails or the USB cannot be read, the message box will pop up as shown below.



SECS/GEM Interface - State



EnableComm：Start Communication

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
10 ： CTLT.GEM尚未執行Communication功能初始化(本項異常不會記錄在Log)

DisableComm：Stop Communication

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
10 ： CTLT.GEM尚未執行Communication功能初始化(本項異常不會記錄在Log)

On Line Request：Send the request to connect with the remote On Line

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
1 ： 目前Control State不是Equipment Off Line State，使用者禁止On Line(本項異常不會記錄於Log)

Off Line：Send the request to disconnect with the remote On Line

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
1 ： 目前Control State不是On Line State或Host Off Line State，使用者禁止Off Line(本項異常不會記錄於Log)

On Line Remote：Switch to remote control status

Part	Description
CGWrapper	CTLT.GEM Control

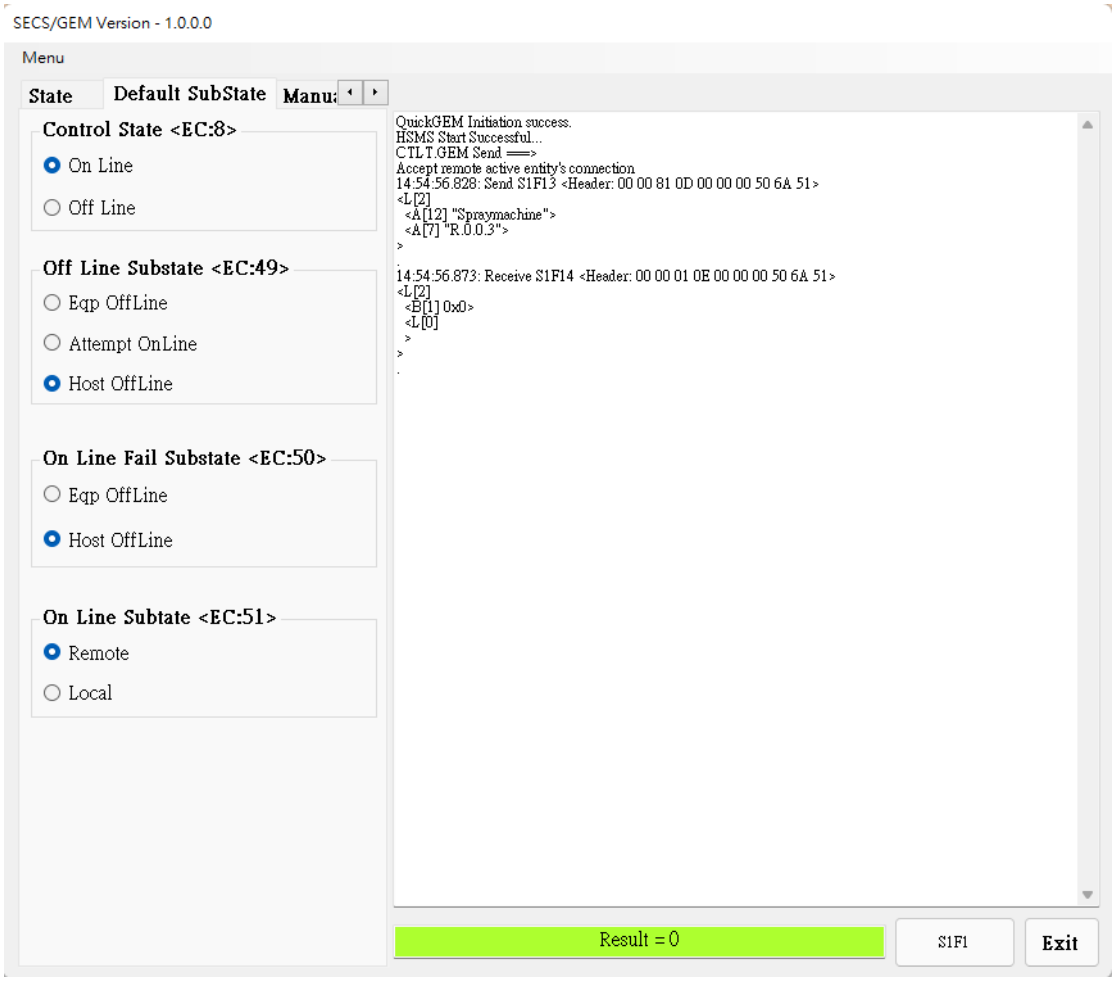
Returns：
0 ： Successful
1 ： 目前Control State不是On Line Local State，故使用者不能要求On Remote Local(本項異常不會記錄於Log)

On Line Local：Switch to local control status

Part	Description
CGWrapper	CTLT.GEM Control

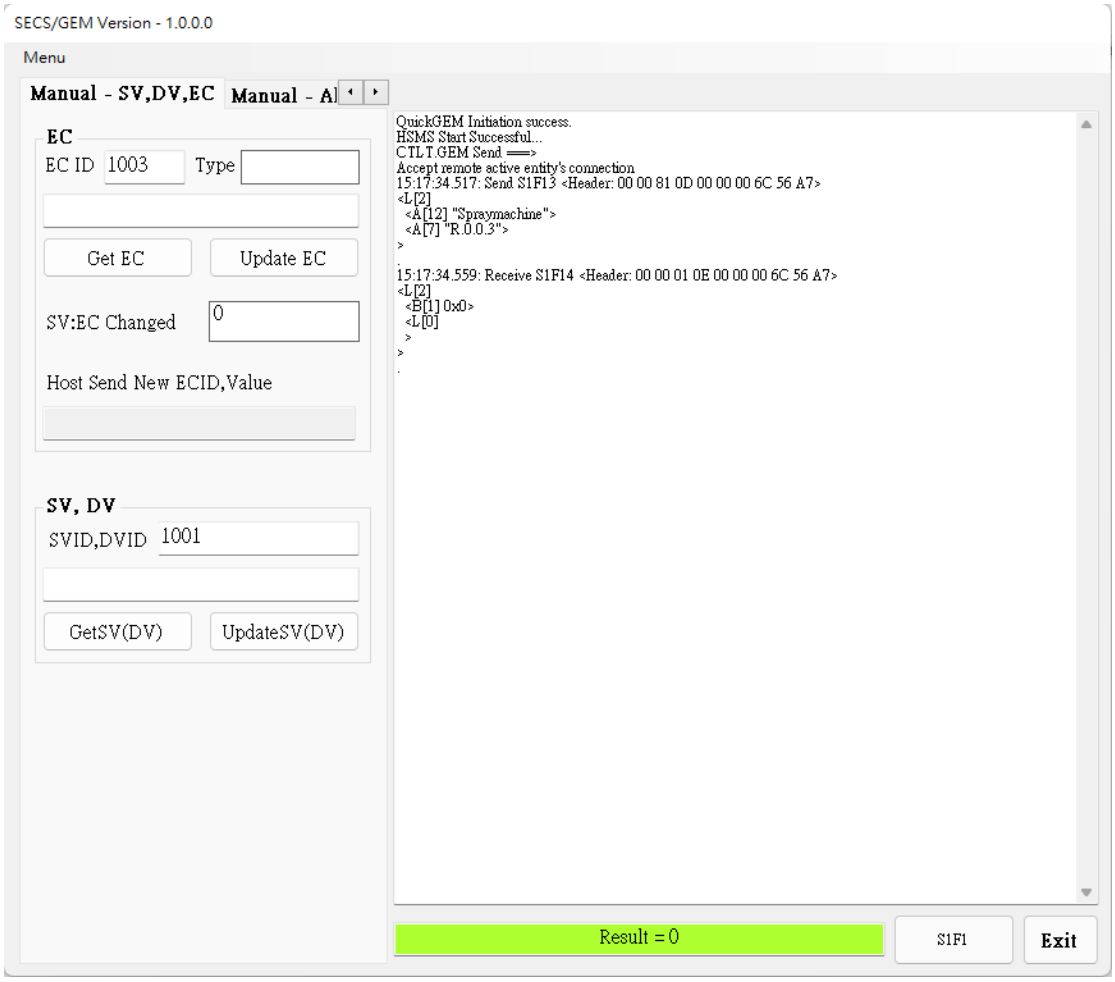
Returns：
0 ： Successful
1 ： 目前Control State不是On Line Remote State，故使用者不能要求On Line Local(本項異常不會記錄於Log)

SECS/GEM Interface – Default Substate



Switch EC:8, EC:49, EC:50, EC:51 ‘s sub state, See [System EC page (p.23)] for details

SECS/GEM Interface – Manual SV,DV,EC



Update EC : Update specified Equipment Constant(EC) value

Part	Description
CGWrapper	CTLT.GEM Control
ECID	[in] Equipment Constant Identifier
Value	[in]Assign value of each equipment constant. The type of value is Variant which means that the value type is assigned depend on Format of ECID defined in initial file.

Returns :
0 : Successful
10 : CTLT.GEM尚未初始化(本項異常不會記錄在Log)
901 : Value之型態無法於COM介面轉換(本項異常不會記錄在Log)

Update SV(DV)：Update specified Status Variable(SV) and Data Value(DV) value

Part	Description
CGWrapper	CTLT.GEM Control
SVID/DVID	[in]Status Variable Identifier / Data Value Identifier
Value	[in]Assign value of each status variable. The type of value is Variant which means that the value type is assigned depend on Format of SVID/DVID defined in initial file.

Returns：
0 ： Successful
10 ： CTLT.GEM尚未初始化(本項異常不會記錄在Log)
901： Value之型態無法於COM介面轉換(本項異常不會記錄在Log)

GetEC：Load specified Equipment Constant value

Part	Description
CGWrapper	CTLT.GEM Control
SVID/DVID	[in]Equipment Constant Identifier
Format	[out]Store the format of each ECID. The format is defined in EC_DATA_TYPE as following： _BINARY_TYPE (33): Binary. _BOOLEAN_TYPE (37): Boolean. _ASCII_TYPE (65): ASCII String. _INT1_TYPE (101): 1 byte integer. _INT2_TYPE (105): 2 byte integer. _INT4_TYPE (113): 4 byte integer. _UINT1_TYPE (165): 1 byte unsigned integer. _UINT2_TYPE (169): 2 byte unsigned integer. _UINT4_TYPE (177): 4 byte unsigned integer. _FT4_TYPE (145): 4 byte floating number. _FT8_TYPE (129): 8 byte floating number.
Value	[out] Store the value of specific ECID.

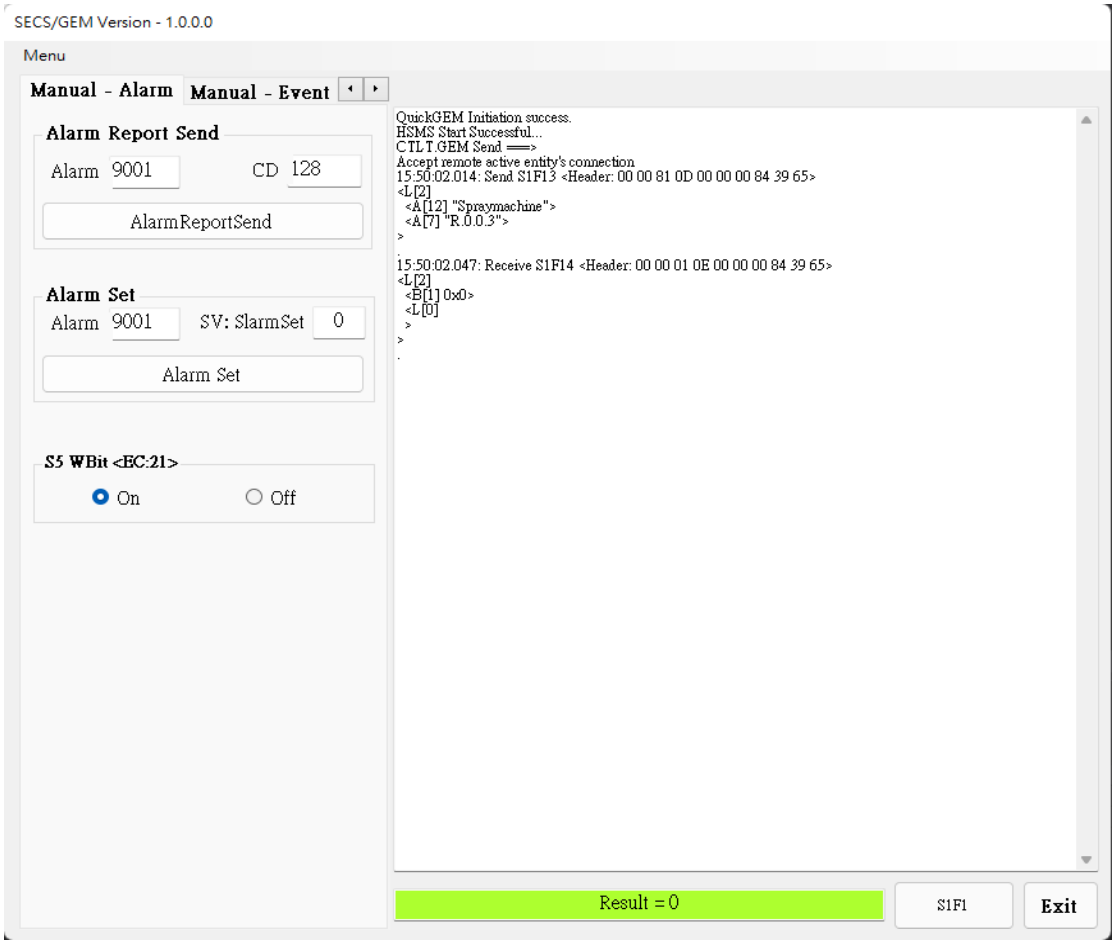
Returns：
0 ： Successful
10 ： CTLT.GEM尚未初始化(本項異常不會記錄在Log)

GetSV(DV)：Load specified Status Variable(SV) and Data Value(DV) value

Part	Description
CGWrapper	CTLT.GEM Control
SVID/DVID	[in]Status Variable Identifier / Data Value Identifier
Format	[out]Store the format of each SVID. The format is defined in SV_DATA_TYPE as following： _BINARY_TYPE (33): Binary. _BOOLEAN_TYPE (37): Boolean. _ASCII_TYPE (65): ASCII String. _INT1_TYPE (101): 1 byte integer. _INT2_TYPE (105): 2 byte integer. _INT4_TYPE (113): 4 byte integer. _UINT1_TYPE (165): 1 byte unsigned integer. _UINT2_TYPE (169): 2 byte unsigned integer. _UINT4_TYPE (177): 4 byte unsigned integer. _FT4_TYPE (145): 4 byte floating number. _FT8_TYPE (129): 8 byte floating number.
Value	[out] Store the value of specific SVID/DVID.

Returns：
0 ： Successful
10 ： CTLT.GEM尚未初始化(本項異常不會記錄在Log)

SECS/GEM Interface – Alarm



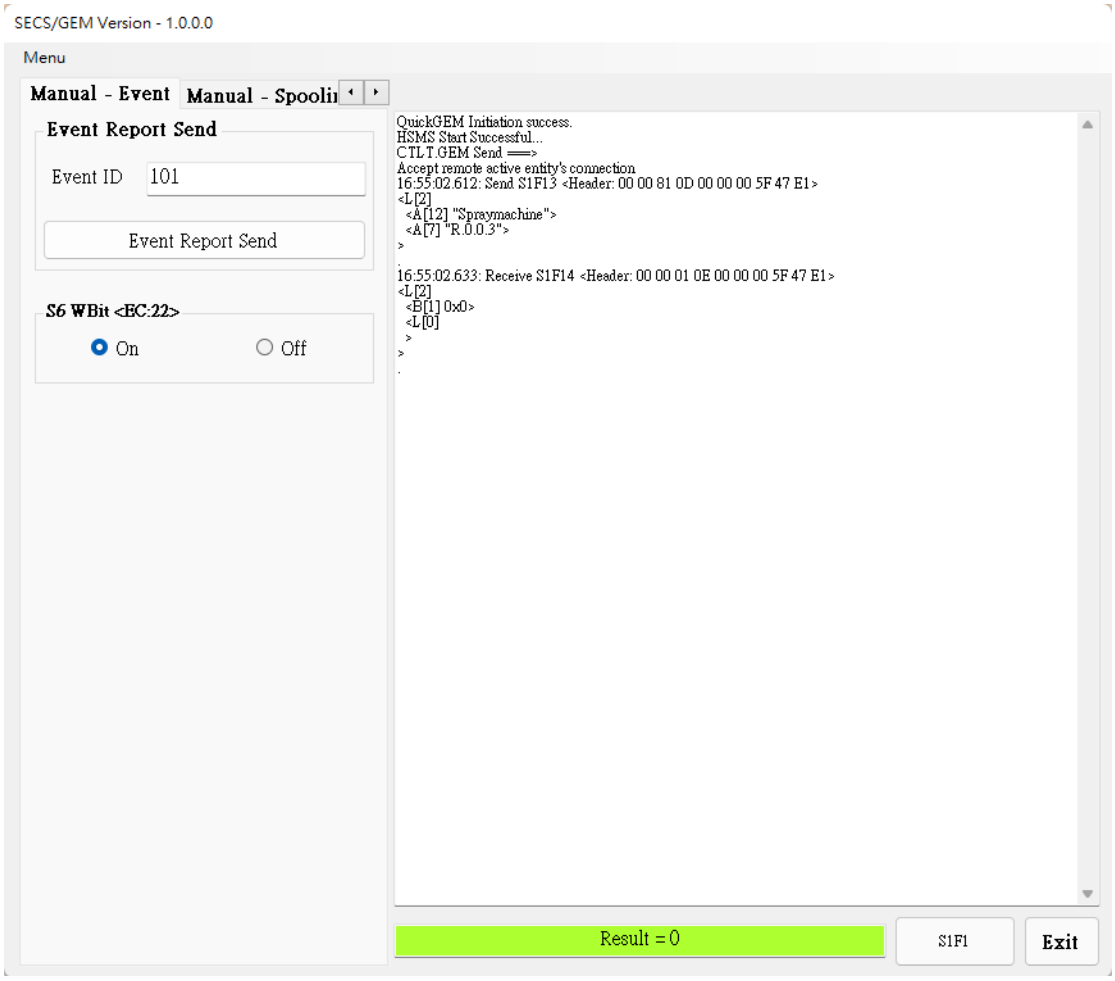
AlarmReportSend：Based on specified Alarm ID send alarm message

Part	Description
CGWrapper	CTLT.GEM Control
AlarmID	[in]Specifies Alarm ID to be sent.
AlarmCD	[in]Specifies Alarm Code to be sent. 依照E30規範AlarmCD，只能使用0以及128(CTLT.GEM並未管制AlarmCD) 0：表示Alarm Reset或結束 128：表示發生Alarm

Returns：
0 ： Successful(訊息成功送出或被Spooled)
1 ： 訊息太大超過CTLT.GEM Buffer容量，故無法送出
3 ： 訊息無法送出，因系統處於Control Off Line State，故禁止送出訊息
4 ： 訊息無法送出，因CLTL.GEM尚未進入Communicating State，故Message禁止送出
5 ： 系統已進入Spool Active State，而S6F11不在Enable Spool List名單內，故不會被Spooled
(以上異常皆不會記錄在Log)

S5 WBit <EC:21> : See [System EC page (p.23)] for details

SECS/GEM Interface – Event



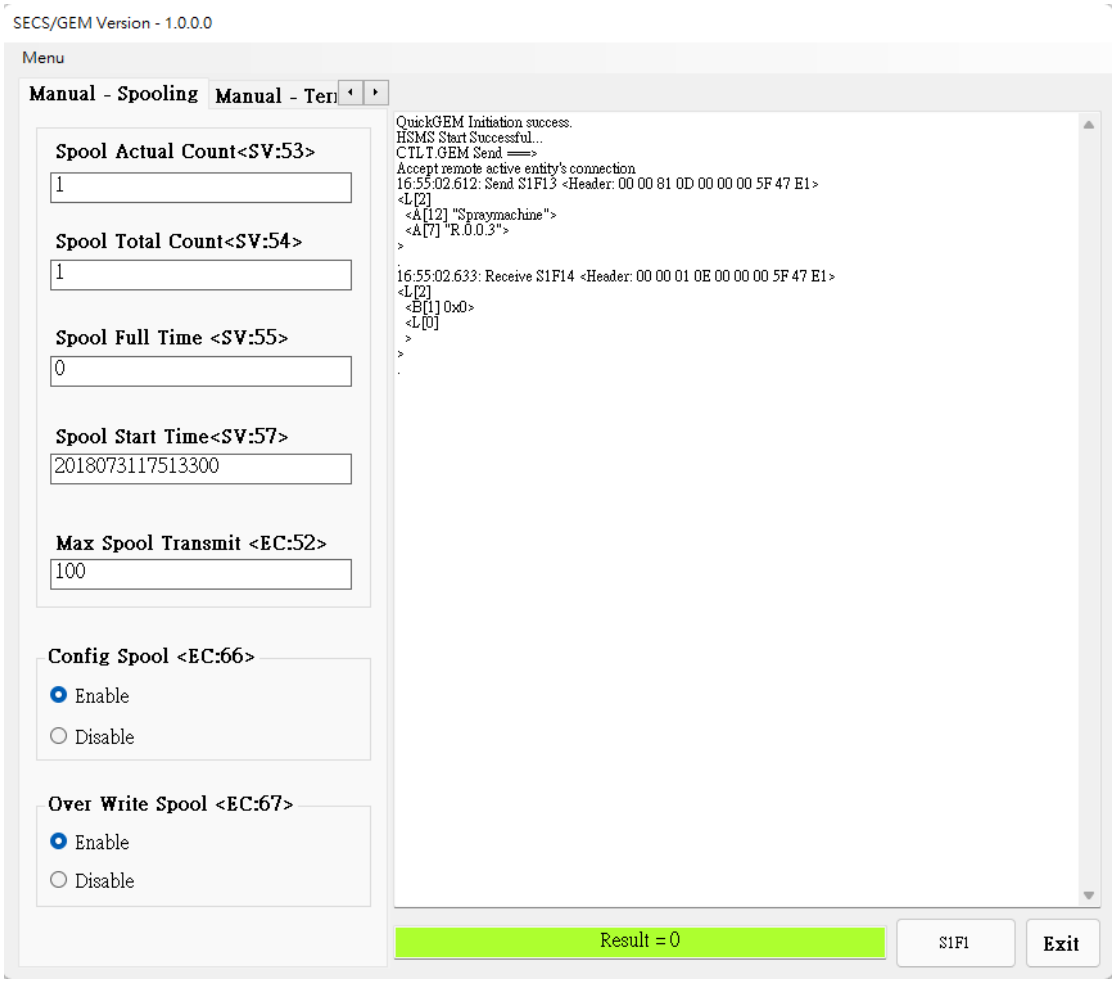
EventReportSend : Based on specified Event ID send event message

Part	Description
CGWrapper	CTLT.GEM Control
EventID	[in]Specifies Event ID to be sent

Returns :
0 : Successful
1 : Event ID找不到，無法送出
3 : 訊息無法送出，因系統處於Control Off Line State，故禁止送出訊息
4 : 訊息無法送出，因CLTL.GEM尚未進入Communicating State，故Message禁止送出
5 : 系統已進入Spool Active State，而S6F11不在Enable Spool List名單內，故不會被Spooled
9 : 該Event ID 被Disable，故無法送出
10 : 訊息太大超過CTLT.GEM Buffer容量，故無法送出
(以上異常皆不會記錄在Log)

S6 WBit <EC:22> : See [System EC page (p.23)] for details

SECS/GEM Interface – Spooling

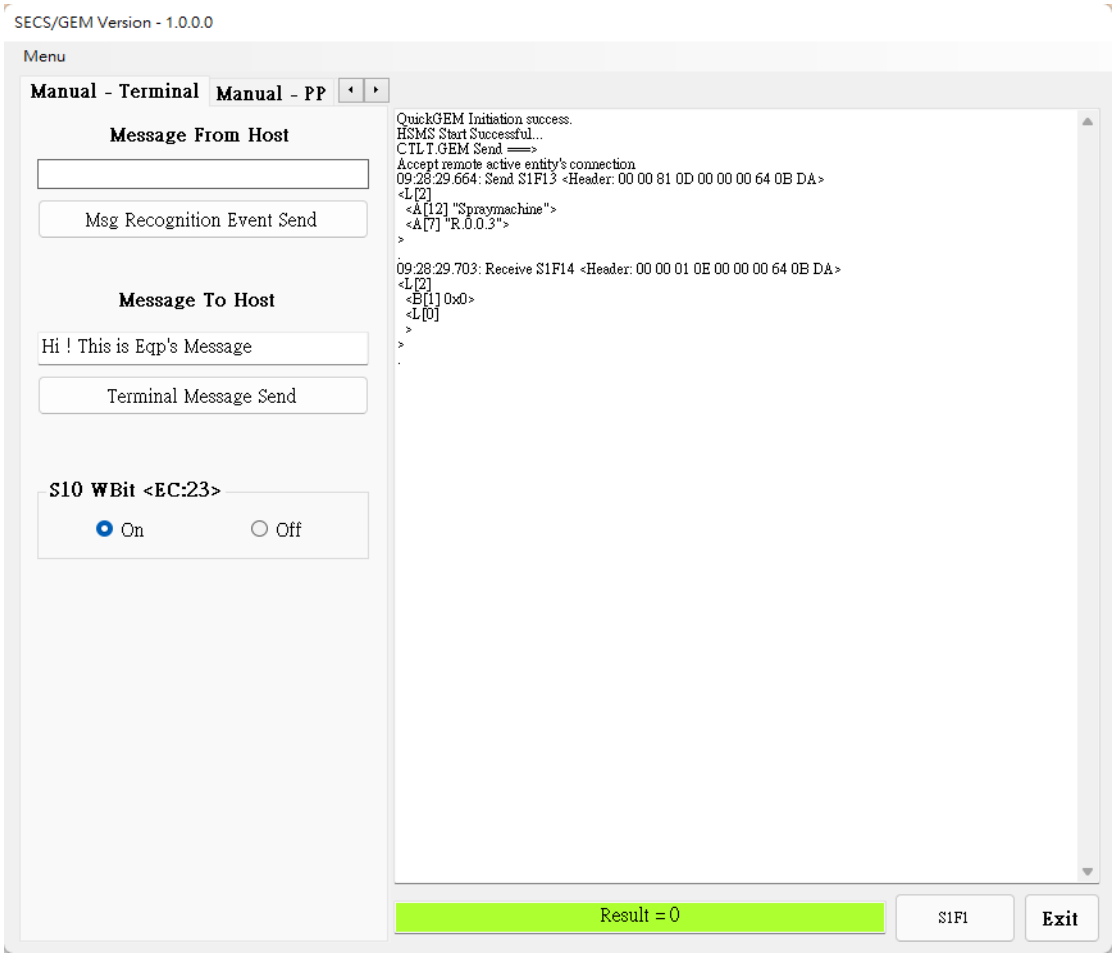


Display SV:53, SV:54, SV:55, SV:57 current value, See [System SV page (p.20)] for details

Display EC:52current value, See [System EC page (p.23)] for details

Open/Close EC66, EC67 enable state, See [System EC page (p.23)] for details

SECS/GEM Interface – Terminal



Msg Recognition Event Send: Send terminal message

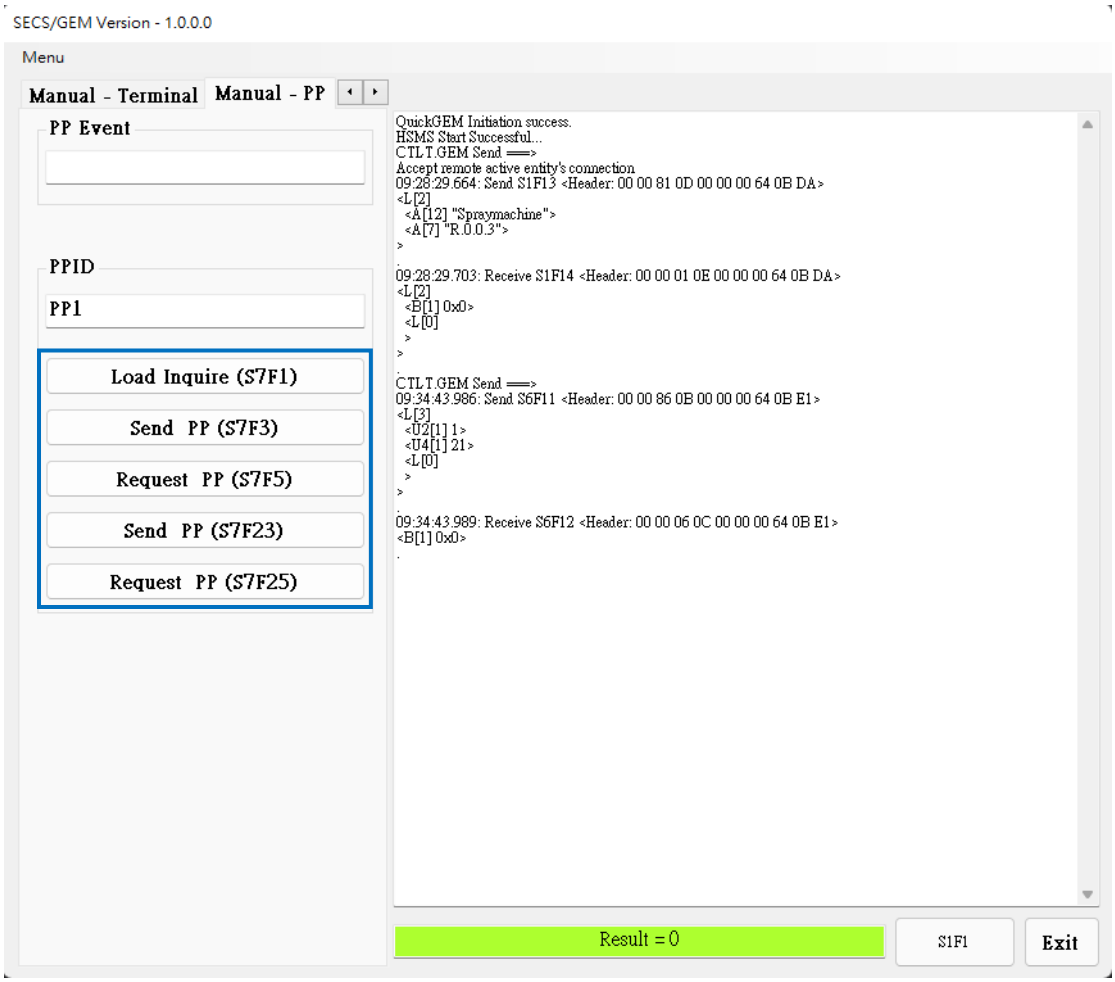
Terminal Message Send: Send terminal message

Part	Description
CGWrapper	CTLT.GEM Control
Message	[in]Specifies Message Text to be sent

Returns :
0 : Successful
10 : CTLT.GEM尚未初始化(本項異常不會記錄在Log)

S10 WBit <EC:23> : See [System EC page (p.23)] for details

SECS/GEM Interface – PP

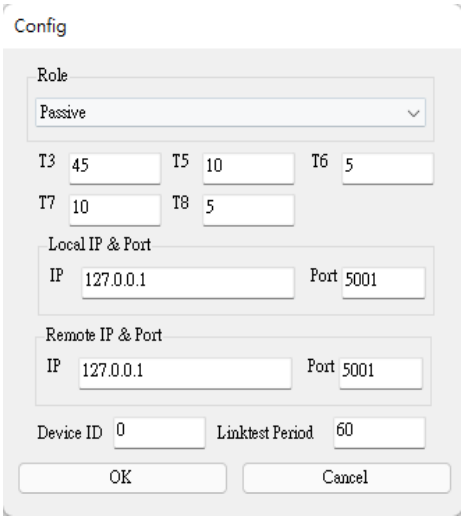


Function specified by excute CmdID

Part	Description
CGWrapper	CTLT.GEM Control
CmdID	[in]命令代號： CMD_PP_LOAD_INQUIRE(11)： Send process program load inquire(S7F1) CMD_UMFORMATTED_PP_SEND(12)： Send process program send(S7F3) CMD_UNFORMATTED_PP_REQUEST(13)： Send process program request(S7F5) CMD_FORMATTED_PP_SEND(14)： Send formatted process program send(S7F23) CMD_FORMATTED_PP_REQUEST(15)： Send formatted process program request(S7F25)
Param1	[in]process program ID(PPID)
Param2	[in]Null

Returns：
1 ： 本機找不到Param1所指定的PPID，無法送出Process Program(for S7F1、S7F3、S7F23)
3 ： 訊息無法送出，因系統處於Control Off Line State，故禁止送出訊息
4 ： 訊息無法送出，因CLTL.GEM尚未進入Communicating State，故Message禁止送出
5 ： 系統已進入Spool Active State，而S6F11不在Enable Spool List名單內，故不會被Spooled
6 ： 本機已存在Param1所指定的PPID且PPOverwriteEnable = No，所以無法送出PP Request訊息(S7F5、S7F25)以要求Host送出Process Program
7 ： CmdID錯誤
10 ： CTLT.GEM尚未執行初始化(本項異常不會記錄在Log)
901： Value型態無法於COM介面轉換(此異常不會記錄在Log)

Config Interface



Role：Set the machine as the active side (Host) or the passive side (Equipment)

T3：Set the time for the sender to wait for the receiver to return a message after sending the data message

T5：After the TCP/IP connection is set to fail, the active side needs to wait for T5 time before reconnecting

T6：Set the time for the sender to wait for the receiver to return a message after sending a control message

T7 : Set the time to wait to enter Selected Stats after entering Connected Stats

T8 : Set the maximum time interval allowed between characters in the message when the receiver receives the message

Local IP & Port : Set local IP & Port

Remote IP & Port : Set remote IP & Port

Device ID : Set Device ID

Linktest Period : Set the interval for sending test messages periodically (limited by T6)

System SV

SVID	Type	Length	SV Name	Remark	說明
1	ASCII	39	License Code		CTLT.GEM Licence的文字說明
2	UINT_1		License Status	0 : No License 1 : Normal License 2 : Small License	CTLT.GEM 授權狀態代碼
3	ASCII	19	GemClock		CTLT.GEM系統時間，格式依照EC : 68 (GemTimeFormat)制定
4	UINT_1		GemControlState	1 = Offline / Equipment Offline 2 = Offline / Attempt Online 3 = Offline / Host Offline 4 = Online Local 5 = Online Remote	目前的Control State
5	UINT_1		GemLinkState	0 : Disable 1 : Enable / Not Communicating 2 : Enable / Communicating	Communicating State
6	UINT_1		SECSCommunicationMode	0 : HSMS Mode(Default) 1 : SECS Mode(Set by APP)	HSMS/SECS-I通訊模式設定，本項預設值為HSMS Mode，若APP使用SECS-I Mode，務必要將本項設定為1，CTLT.GEM才能正常處理Communicating State Model
11	UINT_4		GemPreviousCEID		本項紀錄最後一次所送出的Event ID編號
12	UINT_1		GemOffLineSubState	1 : Eqp Off-Line 2 : Attempt On-Line 3 : Host Off-Line Others : Not at Off-Line	Off-Line的子狀態
13	UINT_1		GemPreviousControlState	1 : Eqp Off-Line 2 : On-Line Local 3 : On-Line Remote	前一次Control State的狀態
14	UINT_1		GemPreviousProcessState	Set by APP	機台前一次的Process State狀態，本項由APP進行Update(本項可刪除)
15	UINT_1		GemProcessState	Set by APP	機台目前的Process State狀態，本項由APP進行Update(本項可刪除)
24	ASCII	6	GemMDLN	Set by APP	機台型號，本項務必使用APP Update

25	ASCII	6	GemSOFTREV	Set by APP	機台軟體，本項務必使用APP Update(本項應以Hard Code方式設計)
39	LIST		AlarmsEnabled		Alarm為Enabled的List資料
40	LIST		AlarmSet		Alarm為Set的List資料
41	LIST		EventsEnabled		Event為Enabled的List資料
42	ASCII	256	GemPPExecName	Set by APP	目前機台所選擇要執行的Process Program Name，本項由APP執行Update(本項可刪除)
43	UINT_1		PPFormate	1：僅支援Unformatted Process Programs 2：僅支援Formatted Process Programs 3：同時支援Unformatted Process Programs和Formatted Process Programs	提供目前的Process Program支援Formatted或Unformatted的功能
53	UINT_4		GemSpoolCountActual		本項記錄目前Spool訊息的實際數量
54	UINT_4		GemSpoolCountTotal		本項紀錄從Spool Active之後，所有被Spool的訊息的數量
55	ASCII	19	GemSpoolFullTime		本項紀錄從Spool進入Full State的時間點
57	ASCII	19	GemSpoolStateTime		本項記錄從Spool進入Active State的時間點
58	UINT_1		GemSpoolState	1：Inactive 2：Active	目前Spool的主要狀態
80	ASCII	20	GemSoftWareRevision		本項為取得CTLT.GEM的軟體版本(Ex：” X.X.X”)

System DV

SVID	Type	Length	SV Name	Remark	說明
9	ASCII	80	GemPPChangeName	Set by APP	當APP有新增、更改、刪除Process Program時，務必要更新本項，以供Host讀取有異動的Process Program名稱
10	UINT_1		GemPPChangeStatus	0 : Created 1 : Changed 2 : Deleted Set by APP	當APP有新增、更改、刪除Process Program時，務必要更新本項，以供Host讀取異動的原因
38	ASCII	19	Alarm ID		最後一次Alarm Report的Alarm ID
46	UINT_1		GemECIDChanged		本項紀錄最後一次被UpdateEC的ECID
47	UINT_1		GemECValueChanged		本項紀錄最後一次被UpdateEC的新EC值
48	UINT_1		GemPreviousECValue		本項紀錄最後一次被UpdateEC的原EC值

System EC

SVID	Type	Min	Max	Default	Unit	EC Name	Remark	說明
7	UINT_1	0	1	0		GemInitCommState	0：Disable 1：Enable	控制Communicating State Model的Default Entry State
8	UINT_1	1	2	1		GemInitControlState	1：Off-Line 2：On-Line	為控制Control State Model的Default Entry State
21	ASCII	0	1	1		GemWBitS5	0：Not Set 1：Set	控制於傳送S5F1時所帶的Wait Bit值
22	UINT_1	0	1	1		GemWBitS6	0：Not Set 1：Set	控制於傳送S6F1時所帶的Wait Bit值
23	UINT_1	0	1	1		GemWBitS10	0：Not Set 1：Set (For S10F1)	控制於傳送S10F1時所帶的Wait Bit值
44	UINT_1	1	10000	5	Sec	GemEtstabCommDelay		控制Communicating State Model 進入Wait Delay Substate的Time Out時間
49	UINT_4	1	3	1		GemOffLineSubState	1：Eqp. Off-Line 2：Attempt On-Line 3：Host Off-Line	控制Control State Model於Off-Line Substate
50	UINT_1	1	3	1		GemOffLineSubState	1：Eqp. Off-Line 3：Host Off-Line	控制Control State Model的Attemp On-Line無法On-Line時要進入的Off-Line Substate
51	UINT_1	4	5	4		GemPreviousControlState	4：On-Line / Local 5：On-Line / Remote	控制Control State Model於On-Line Substate的Default Entry State
52	UINT_1	1	2E+08	100		GemMaxSpoolTransmit		控制Spool State Model的Transmit Spool SubState一次最多可以傳送的訊息數量。當送達此數量，並且Spool尚有未傳送的訊息時，則Spool State Model會回到NO SPOOL OUTPUT Substate

66	UINT_1	0	1	0		GemConfigSpool	0：Disable 1：Enable	控制Spooling功能是否要啟動。 注意：依照SEMI E30的規範，若系統已進入Spooling Active State後再Disable Spooling的功能，並不會改變Spooling目前於SPOOL ACTIVE STATE的狀態
67	ASCII	0	1	0		GemOverWriteSpool	0：Do not overwrite 1：Overwrite	當Spool的空間滿了之後，若有新的訊息要加入到Spool時，是否能夠取代Spool中舊有的訊息。 注意：若設定Do not overwrite，新訊息將會被捨棄
68	ASCII	0	3	1		GemTimeFormat	0：12-bytes 1：16-bytes 2：14-bytes 3：ISO8601 format	設定”時間”的傳輸格式
71	LIST	1	6	5		GemDATAIDFormat	1：INT_1 2：INT_2 3：INT_4 4：UINT_1 5：UINT_2 6：UINT_4	送S6F11時，設定DATAID之格式。本項為CTLT.GEM自動產生的流水號值，範圍於0~65535之間
75	LIST	1	6	5		GemSAMPLEFormat	1：INT_1 2：INT_2 3：INT_4 4：UINT_1 5：UINT_2 6：UINT_4	Trace data功能中，送S6F1訊息時，設定訊息SAMPLN item的格式

System Event

SVID	Event Name	說明
3	GemProcessProgramChange (Set by APP)	當機台Process State有變動時，必須要報Event給Host。本Event由APP負責送出，建議APP應該用獨立不同的Event告知Host目前的Process State (本項可刪除)
8	GemControlStateLocal	當GEM的Control State Model進入On-Line Local Substate時，核心會送出本訊息
9	GemControlStateRemote	當GEM的Control State Model進入On-Line Remote Substate時，核心會送出本訊息
20	GemEqConstChanged	當APP Call UpdateEC函式時(使用者修改EC)，核心會送出本訊息 註：當Host以S2F15改變設備EC時，核心並不會送出本訊息
21	GemMessageRecognition (Set by APP)	當APP收到S10F3時，Operator應該按下「訊息確認」的按鍵，送出確認的Event，已告知Host，Operator已閱讀訊息，本Event由APP負責送出。若Host送出多個S10F3，Operator只需按一次確認鍵，表示已讀取所有訊息
22	GemEqpOffLine	當GEM的Control State Model進Off-Line State時，核心會送出本訊息
23	GemSpoolingActived	當GEM的Spool State Model進入SPOOL ACTIVE State時，核心會產生本訊息並且存到Spool中。因此GemSpoolingActived的Event一定會是第一個被Spool的訊息
24	GemSpoolingDeactived	當GEM Spool中的訊息被送完或被清除時，核心會送出本訊息
25	GemSpoolTransmitFailure	當GEM於TRANSMIT SPOOL Substate傳送訊息時，若發生Communicating State Model進入NOT COMMUNICATING State則於重新進入COMMUNICATING State時，核心會送出本訊息 注意：本Event非SEMI E30規範中必須送的Event，可以設定本Event為Disable，但請勿刪除本項Event

Equpiment SV

ID	TYPE	Lenght	NAME	Definition
1001	ASCII	1000	Equipment_RemoteControlStatus	遠端控制指令
1002	ASCII	1000	Equipment_RemoteControlResult	遠端控制指令執行結果
1003	ASCII	1000	Equipment_LocalControlStatus	設備端控制指令
1004	ASCII	1000	Equipment_LocalControlResult	設備端控制指令結果
1005	ASCII	1000	Reel_Status	Reel 狀態
1006	ASCII	1000	DieCount_CompareResult	Die Count 比對結果

Equpiment DV

ID	TYPE	Length	NAME	Definition
4001	ASCII	1000	ReelID	ReelID
4002	ASCII	1000	APN	APN
4003	ASCII	1000	DateCode	DateCode
4004	ASCII	1000	Lot6	Lot6
4005	ASCII	1000	CurrentRecipeName	當前參數名稱
4006	ASCII	1000	Expected_DieCount	預期 Die Count 數量
4007	ASCII	1000	Actual_DieCount	實際 Die Count 數量

Equpiment Event

ID	NAME	Definition
101	ReelID_Send	傳送 ReelID 事件
102	Recipe_Change	更換參數檔事件
103	Equipment_RemoteControl	遠端控制事件
104	Equipment_LocalControl	設備端控制事件
105	EndLot	結批事件(Die Count Send)
106	Reel_CurrentStatus	Reel 安裝/移除事件

PP BODY

[AxisAll]

NAME	Definition
Axis1Pos0	左馬達_原點位置
Axis1Pos0Speed	左馬達_初始速度
Axis1Pos1	左馬達_移動距離
Axis1Pos1Speed	左馬達_速度
Axis2Pos0	載帶馬達_原點位置
Axis2Pos0Speed	載帶馬達_初始速度
Axis2Pos1	載帶馬達_移動距離
Axis2Pos1Speed	載帶馬達_速度
Axis2Pos1SpeedBin	載帶馬達_速度 Bin 值
Axis3Pos0	右馬達_原點位置
Axis3Pos0Speed	右馬達_初始速度
Axis3Pos1	右馬達_移動距離
Axis3Pos1Speed	右馬達_速度
Axis4Pos0	軸 4_原點位置

Axis4Pos0Speed	軸 4_初始速度
Axis4Pos1	軸 4_移動距離
Axis4Pos1Speed	軸 4_速度
Axis1JogSpeed	左馬達_吋動速度
Axis2JogSpeed	載帶馬達_吋動速度
Axis3JogSpeed	右馬達_吋動速度
Axis4JogSpeed	軸 4_吋動速度
Axis1DataSet	左馬達_數據集
Axis2DataSet	載帶馬達_數據集
Axis2DataSet28800	載帶馬達_數據集 2
Axis3DataSet	右馬達_數據集
Axis4DataSet	軸 4_數據集

[OtherPar]

NAME	Definition
Stand1NpTime	載帶整定時間
IndexBSet	A-B 距離設定(次數)
NgNpTime	NgNp 次數
NozzleErrorSet	吸嘴設定
YieldErrorSet	良率錯誤設定
CcdTrgTime	CCD 觸發逾時
Ccd6SpaceSet	CCD6 設定值
LoadHot1Time	加溫器加溫時間
LoadHot2TimeDown	加溫器 2 降溫時間
LoadHot2TimeUp	加溫器 2 加溫時間
LoadReelStopTime	載入停止時間
LoadCaseStopTime	載入停止時間
LoadCCD6DelayTime	CCD6 延遲時間
LoadHotDelayTime	加溫訊號延遲
LoadOutTime	載入時間計時
Axis1AccDec	Axis1 馬達加速度

Axis2AccDec	Axis2 馬達加速度
Axis3AccDec	Axis3 馬達加速度
IndexCSet	數量到達後移動設定 (次數)
Light1	A 站上側光源
Light2	A 站下側光源
Light3	B 站上側光源
Light4	B 站下側光源
Light5	C 站光源
Light6	A 站底部光源
Light7	B 站底部光源
Light8	C 站底部光源

Event Report Link

Equipment event (p.28) will be transmitted with RTPID and SV, DV, EC values will be transmitted together, as shown below.

EventID	RPTID	SVID, DVID, ECID
ReelID_Send (101)	101	ReelID (4001)
Recipe_Change (102)	102	CurrentRecipeName (4005)
Equipment_RemoteControl (103)	103	Equipment_RemoteControlStatus (1001) Equipment_RemoteControlResult (1002)
Equipment_LocalControl (104)	104	Equipment_LocalControlStatus (1003) Equipment_LocalControlResult (1004)
EndLot (105)	105	Actual_DieCount (4007)
Reel_CurrentStatus (106)	106	Reel_Status (1005)

SECS-II Message List (H -> E)

Host → Equipment (Primary)		Equipment → Host (Secondary)	
Sn Fn		Sn F0	Abort Transaction
S1 F1	Are You There Request	S1 F2	On Line Data
S1 F3	Selected Equipment Status Request	S1 F4	Selected Equipment Status Data
S1 F11	Status Variable Name List Request	S1 F12	Status Variable name List Reply
S1 F13	Establish Communications Request	S1 F14	Establish Communications Request Acknowledge
S1 F15	Request Off-Line	S1 F16	Off-Line Acknowledge
S1 F17	Request On-Line	S1 F18	On-Line Acknowledge
S2 F13	Equipment Constant Request	S2 F14	Equipment Constant Data
S2 F15	New Equipment Constant Send	S2 F16	New Equipment Constant Acknowledge
S2 F17	Date and Time Request	S2 F18	Date and Time Data
S2 F23	Trace Initialize Send	S2 F24	Trace Initialize Acknowledge
S2 F25	Loop Back Diagnosis Request	S2 F26	Loop Back Diagnosis Reply
S2 F29	Equipment Constant Name List Request	S2 F30	Equipment Constant Name List Reply
S2 F31	Date and Time Send	S2 F32	Date and Time Acknowledge
S2 F33	Define Report	S2 F34	Define-Report Acknowledge
S2 F35	Link Event Report	S2 F36	Link Event Report Acknowledge
S2 F37	Enable/Disable Event Report	S2 F38	Enable/Disable Event Report Acknowledge
S2 F41	Remote Command Send	S2 F42	Remote Command Acknowledge
S5 F3	Enable/Disable Alarm Send	S5 F4	Enable/Disable Alarm Acknowledge
S5 F5	List Alarms Request	S5 F6	List Alarm Data
S5 F7	List Enabled Alarms Request	S5 F8	List Enabled Alarm Data
S6 F15	Event Report Request	S6 F16	Event Report Data
S6 F19	Individual Report Request	S6 F20	Individual Report Data
S7F1	Process Program Load Inquire	S7F2	Process Program Load Grant
S7F5	Process Program Request	S7F6	Process Program Data
S7F19	Current Process Program Dir. Request	S7F20	List Process Program Data
S7F25	Formatted Process Program Request	S7F26	Formatted Process Program Data
S10 F3	Terminal Display	S10 F4	Terminal Display Acknowledge

SECS-II Message List (E -> H)

Host → Equipment (Secondary)		Equipment → Host (Primary)	
Sn Fn		Sn F0	Abort Transaction
S1 F2	On Line Data	S1 F1	Are You There Request
S1 F14	Establish Communications Request Acknowledge	S1 F13	Establish Communications Request
S2 F18	Date and Time Data	S2 F17	Date and Time Request
S5 F2	Alarm Report Acknowledge	S5 F1	Alarm Report Send
S6 F2	Trace Data Acknowledge	S6 F1	Trace Data Send
S6F12	Event Report Acknowledge	S6F11	Event Report Send
		S9 F7	Illegal Data
		S9 F9	Transaction Timer Timeout
S10 F2	Terminal Request Acknowledge	S10 F1	Terminal Request

SECS-II Message Example (Host perspective)

S1F1 : Are You There (H -> E)

Description **:**

The host may send this message to the EQ at any time. EQ sends this message periodically as a “heartbeat” to determine if the SECS link is alive.

```
2022/2/24 10:06:33.632: Send S1F1 <Header: 00 00 81 01 00 00 00 02 B0 E2>
<L[1]
 <A[14] "Are you there?">
>
.
```

S1F2 : On-Line Data (E -> H)

Description **:**

EQ reports its Model Number and Software Revision to Host

```
2022/2/24 10:06:33.633: Receive S1F2 <Header: 00 00 01 02 00 00 00 02 B0 E2>
<L[2]
 <A[13] "SMT Equipment">
 <A[7] "1.0.0.0">
>
.
```

.

S1F3 : Selected Status Request (H -> E)

Description :

A request to EQ to report selected values of its status.

```
2022/2/24 10:08:19.665: Send S1F3 <Header: 00 00 81 03 00 00 00 02 B0 ED>
<U4[3] 100001 2002 2003>
.
```

S1F4 : Selected Status Data (E -> H)

Description :

The equipment reports the value of each SVID requested in the order requested.

```
2022/2/24 10:08:19.665: Receive S1F4 <Header: 00 00 01 04 00 00 00 02 B0 ED>
<L[3]
 <U1[1] 0>
 <A[0] "">
 <F4[1] 0>
>
.
```

S1F11 : Status Variable NamelistRequest (H -> E)

Description :

A request to the equipment to identify certain status variables.

```
2022/2/24 10:10:52.536: Send S1F11 <Header: 00 00 81 0B 00 00 00 02 B0 EF>
<L[2]
 <U4[1] 2002>
 <U4[1] 2003>
>
.
```

S1F12 : Status Variable Namelist Reply (E -> H)

Description :

EQ reports to the Host the name and units of the requested SVs.

```
2022/2/24 10:10:52.538: Receive S1F12 <Header: 00 00 01 0C 00 00 00 02 B0 EF>
<L[2]
 <L[3]
  <U4[1] 2002>
  <A[19] "SV_CONTROL_JOB_NAME">
  <A[0] "">
 >
 <L[3]
  <U4[1] 2003>
  <A[17] "SV_INMP1_NOW_TEMP">
  <A[6] "Degree">
 >
>
.
```

S1F13 : Establish Communication Request (H -> E)

Description:
Either end of the link may send S1F13 as the first message to establish connection of the link. Under EQ's initialization logic, EQ will always attempt to establish the link. If no Host acceptance reply is received. EQ sends this message periodically.

```
2022/2/24 10:12:02.624: Send S1F13 <Header: 00 00 81 0D 00 00 00 02 B0 F0>
<L[0]
>
.
```

S1F14 : Establish Communication Request Acknowledge (E -> H)

Description:
Accept or deny Establish Communication Request.

```
2022/2/24 10:12:02.625: Receive S1F14 <Header: 00 00 01 0E 00 00 00 02 B0 F0>
<L[2]
 <B[1] 0x00>
 <L[2]
  <A[13] "SMT Equipment">
  <A[7] "1.0.0.0">
 >
>
.
```

S1F15 : Request Off-Line (H -> E)

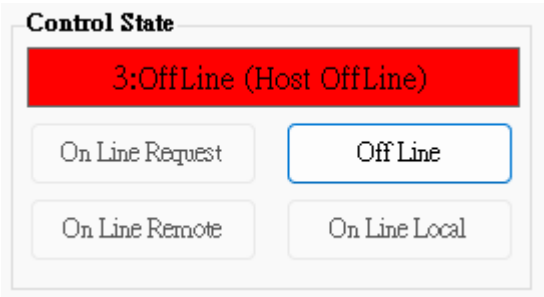
Description:
Host requests EQ to go Off-Line

```
2022/2/24 10:12:48.521: Send S1F15 <Header: 00 00 81 0F 00 00 00 02 B0 F1>
.
```

S1F16 : Off-Line Acknowledge (E -> H)

Description:
EQ control state transits to Host Off-Line and sends this message in response to S1F15 primary.

```
2022/2/24 10:12:48.522: Receive S1F16 <Header: 00 00 01 10 00 00 00 02 B0 F1>
<B[1] 0x00>
.
```



S1F17 : Request On-Line (H -> E)

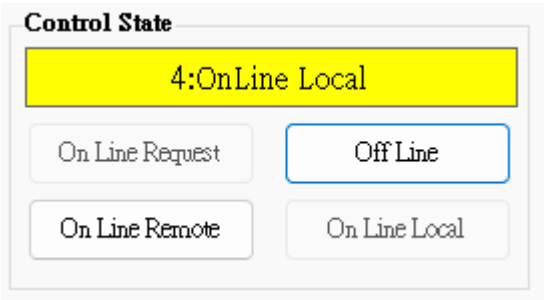
Description:
Host requests EQ to go On-Line

```
2022/2/24 10:14:33.960: Send S1F17 <Header: 00 00 81 11 00 00 00 02 B0 F2>
.
```

S1F18 : On-Line Acknowledge (E -> H)

Description:
EQ responds to S1F17 primary in an attempt to go On-Line. Data item ONLACK indicates the success of failure of the attempt.

```
2022/2/24 10:14:33.961: Receive S1F18 <Header: 00 00 01 12 00 00 00 02 B0 F2>
<B[1] 0x00>
.
```



S2F13 : Equipment Constant Request (H -> E)

Description:
Constants and other values that are changed infrequently can be obtained using this message.

```
2022/2/24 10:04:22.056: Send S2F13 <Header: 00 00 82 0D 00 00 00 02 B0 E0>
<L[3]
 <U4[1] 1001>
 <U4[1] 1002>
 <U4[1] 1003>
>
.
```

S2F14 : Equipment Constant Data (E -> H)

Description:
EQ returns constant data in the order requested in S2F13.

```
2022/2/24 10:04:22.057: Receive S2F14 <Header: 00 00 02 0E 00 00 00 02 B0 E0>
<L[3]
 <U1[1] 1>
 <U2[1] 50>
 <F4[1] 3.456>
>
.
```

S2F15 : New Equipment Constant Send (H -> E)

Description:
Change one or more equipment constants.

```
2022/2/24 10:02:43.540: Send S2F15 <Header: 00 00 82 0F 00 00 00 02 B0 DF>
<L[3]
 <L[2]
  <U4[1] 1002>
  <U2[1] 50>
 >
 <L[2]
  <U4[1] 1003>
  <F4[1] 3.456>
 >
 <L[2]
  <U4[1] 1011>
  <A[40] "_123456789_123456789_123456789_123456789">
 >
>
.
```

S2F16 : Equipment Constant Send Acknowledge (E -> H)

Description:
Change one or more equipment constants.

```
2022/2/24 10:02:43.581: Receive S2F16 <Header: 00 00 02 10 00 00 00 02 B0 DF>
<B[1] 0x00>
.
```

EAC	Description
0x00	OK.
0x01	At least one ECID is invalid.
0x03	At least one ECV is out of range.

S2F17 : Date and Time Request (H -> E)

Description:
Host sends this message at any time to determine the date and time base EQ is currently using.

```
2022/2/24 10:15:34.543: Send S2F17 <Header: 00 00 82 11 00 00 00 02 B0 F4>
.
```

S2F18 : Date and Time Data (E -> H)

Description:
Host sends this message at any time to determine the date and time base EQ is currently using.

```
2022/2/24 10:15:34.543: Send S2F17 <Header: 00 00 82 11 00 00 00 02 B0 F4>
.
2022/2/24 10:15:34.544: Receive S2F18 <Header: 00 00 02 12 00 00 00 02 B0 F4>
<A[16] "2022022410153400">
.
```

S2F23 : Trace Initialize Send (H -> E)

Description:
Trace initial send to device to enable/disable trace initial data. The trace data is returned on (S6F1) and is related to the original request by the TRID.

```
2022/2/24 10:35:34.704: Send S2F23 <Header: 00 00 82 17 00 00 00 02 B0 F5>
<L[5]
 <U4[1] 123456789>
 <A[6] "000003">
 <U4[1] 27>
 <U4[1] 3>
 <L[3]
  <U4[1] 100001>
  <U4[1] 2002>
  <U4[1] 2003>
 >
>
.
```

S2F24 : Trace Initialize Acknowledge (E -> H)

Description:
Acknowledge or error for Trace Initialize Send(S2F23)

```
2022/2/24 10:35:34.707: Receive S2F24 <Header: 00 00 02 18 00 00 00 02 B0 F5>
<B[1] 0x00>
.
```

TIAACK	Description
0x00	Normal. Everything correct.
0x03	Invalid period. DSPER is incorrect. The Equipment does not start the trace.

S2F25 : Diagnostic Loopback Request (H -> E)

Description:
Host sends an arbitrary binary string to test operation of the link.

```
2022/2/24 10:41:10.503: Send S2F25 <Header: 00 00 82 19 00 00 00 02 B0 F6>
<B[10] 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09>
.
```

S2F26 Diagnostic Loopback Data (E -> H)

Description:
EQ “echoes” back the binary data from Host’s S2F25.

```
2022/2/24 10:41:10.504: Receive S2F26 <Header: 00 00 02 1A 00 00 00 02 B0 F6>
<B[10] 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09>
.
```

S2F29 Equipment Constant Namelist Request (H -> E)

Description:
This function allows the host to retrieve basic information about what equipment constants are available in the equipment.

```
2022/2/24 10:46:20.590: Send S2F29 <Header: 00 00 82 1D 00 00 00 02 B0 F8>
<L[2]
 <U4[1] 1001>
 <U4[1] 1003>
>
.
```

S2F30 : Equipment Constant Namelist Reply (E -> H)

Description:
Data Response for Equipment Constant Namelist Request(S2F29).

```
2022/2/24 10:46:20.590: Receive S2F30 <Header: 00 00 02 1E 00 00 00 02 B0 F8>
<L[2]
 <L[6]
  <U4[1] 1001>
  <A[10] "EC1_UINT_1">
  <A[1] "0">
  <A[1] "1">
  <A[1] "1">
  <A[0] "">
 >
 <L[6]
  <U4[1] 1003>
  <A[8] "EC3_FT_4">
  <A[3] "0.1">
  <A[4] "10.9">
  <A[5] "3.456">
  <A[6] "Degree">
 >
>
.
```

S2F31 : Date and Time Send (H -> E)

Description:
Useful to synchronize the equipment time with the host time base.

```
2022/2/24 10:52:10.325: Send S2F31 <Header: 00 00 82 1F 00 00 00 02 B1 06>
<A[16] "2013081711223300">
.
```

S2F32 Date and Time Acknowledge (E -> H)

Description:
Acknowledge the receipt of time and date.

```
2022/2/24 10:52:10.326: Receive S2F32 <Header: 00 00 02 20 00 00 00 02 B1 06>
<B[1] 0x01>
.
```

TIACK	Description
0x00	Normal. Everything correct.
0x01	Invalid Date and/or Time.

S2F33 Define Report (H -> E)

Description:
The purpose of this message is for the host to define a group of reports for the equipment. The event report will be sent via S6F11 when the event occurred.

```
2022/2/24 10:55:47.918: Send S2F33 <Header: 00 00 82 21 00 00 00 02 B1 08>
<L[2]
 <U1[1] 0>
 <L[1]
  <L[2]
   <U4[1] 7001>
   <L[3]
    <U4[1] 17>
    <U4[1] 100001>
    <U4[1] 2002>
   >
  >
 >
>
.
```

S2F34 Define Report Acknowledge (E -> H)

Description:
Acknowledge or error if an error condition is detected the entire message is rejected(i.e., partial changes are not allowed)

```
2022/2/24 10:55:47.932: Receive S2F34 <Header: 00 00 02 22 00 00 00 02 B1 08>
<B[1] 0x00>
.
```

S2F35 : Link Event Report (H -> E)

Description:
The purpose of this message is for the host to link n reports to an event(CEID). These linked event reports will default to 'disabled' upon linking. That is, the occurrence of an event would not cause the report to be sent until enabled. See S2F37 for enabling reports.

```
2022/2/24 11:00:04.566: Send S2F35 <Header: 00 00 82 23 00 00 00 02 B1 09>
<L[2]
 <U1[1] 0>
 <L[1]
  <L[2]
   <U4[1] 101>
   <L[1]
    <U4[1] 7001>
   >
  >
 >
>
.
```

S2F36 Link Event Report Acknowledge (E -> H)

Description:
Acknowledge or error if an error condition is detected the entire message is rejected(i.e., partial changes are not allowed).

```
2022/2/24 11:00:04.576: Receive S2F36 <Header: 00 00 02 24 00 00 00 02 B1 09>
<B[1] 0x00>
.
```

S2F37 : Enable/Disable Event Report (H -> E)

Description:
The purpose of this message is for the host to enable or disable reporting for a group of events(CEIDs).

```
2022/2/24 11:02:00.886: Send S2F37 <Header: 00 00 82 25 00 00 00 02 B1 0A>
<L[2]
 <Boolean[1] 0x1>
 <L[2]
  <U4[1] 101>
  <U4[1] 102>
 >
>
.
```

S2F38 : Enable/Disable Event Report Acknowledge (E -> H)

Description:
Acknowledge or error if an error condition is detected the entire message is rejected(i.e., partial changes are not allowed).

```
2022/2/24 11:02:00.887: Receive S2F38 <Header: 00 00 02 26 00 00 00 02 B1 0A>
<B[1] 0x00>
.
```

ERACK	Description
0x00	OK.
0x01	Denied. At least one CEID does not exist.

S2F41 : Remote Command Send (H -> E)

Description:
The host requests the equipment perform the specified remote command with the associated parameters.

```
2022/2/24 13:54:12.874: Send S2F41 <Header: 00 00 82 29 00 00 00 6D 0A 29>
<L[2]
 <A[9] "PP-SELECT">
 <L[2]
  <L[2]
   <A[4] "PPID">
   <A[3] "PP1">
  >
  <L[2]
   <A[6] "PortID">
   <U1[1] 1>
  >
 >
>
.
```

RCMD String	Function
"START"	Starts a Process Program.
"ABORT"	Aborts a running Process Program.
"PPSELECT"	Pauses a running Process Program.
" LOCAL "	On-Line/LOCAL
" REMOTE "	On-Line/REMOTE

S2F42 : Remote Command Acknowledge (E -> H)

Description:

Acknowledge host's command or error. If command is not accepted due to one or more invalid parameters, then a list of invalid parameters will be returned containing the parameter name and reason for being invalid.

```
2022/2/24 13:54:12.875: Receive S2F42 <Header: 00 00 02 2A 00 00 00 6D 0A 29>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

HCACK	Description
0x00	OK. All normal.
0x01	Invalid command. Equipment rejects command.
0x02	Cannot perform now. Equipment rejects command.
0x03	At least one parameter is invalid. Equipment rejects the command.
0x40	Control State is Local. Equipment rejects command.
0x41	Incorrect Process State for this command. Equipment rejects command.
0x42	Process Program contains an invalid format. Unable to load program.

S5F1 : Alarm Report (E -> H)

Description:
This message reports a change in or presence of an alarm condition. One message will be issued when the alam is set and one message will be issued when the alarm is cleared. Irrecoverable errors and attention flags may not have a corresponding clear message.

```
2022/2/24 15:58:45.159: Receive S5F1 <Header: 00 00 85 01 00 00 00 97 D9 75>
<L[3]
 <B[1] 0x80>
 <U4[1] 9001>
 <A[17] "Pump Press Fressl">
>
.
```

S5F2 : Alarm Acknowledge (H -> E)

Description:
The host acknowledges the alarm report. This equipment ignores the ACKC5.

```
2022/2/24 15:58:45.162: Send S5F2 <Header: 00 00 05 02 00 00 00 97 D9 75>
<B[1] 0x00>
.
```

S5F3 Enable/Disable Alarm Send (H -> E)

Description:
This message will change the state of the enable bit in the equipment. The enable bit determines if the alarm will be sent to the host. Alarms which are not controllable in this way are unaffected by this message.

```
2022/2/24 14:19:08.561: Send S5F3 <Header: 00 00 85 03 00 00 00 48 85 D9>
<L[2]
 <B[1] 0x80>
 <U4[1] 9001>
>
.
```

ALED	Description
0x80	Enable sending alarm.
0x00	Disable sending alarm.

S5F4 Enable/Disable Alarm Acknowledge (E -> H)

Description:
The host acknowledges the alarm report. This equipment ignores the ACKC5.

```
2022/2/24 14:19:08.565: Receive S5F4 <Header: 00 00 05 04 00 00 00 48 85 D9>
<B[1] 0x00>
.
```

ACKC5	Description
0x00	Normal. Everything correct.
0x01	Invalid ALID.

S5F5 : List Alarms Request (H -> E)

Description:
This message requests the equipment to send binary and analog alarm information to the host.

```
2022/2/24 14:23:57.458: Send S5F5 <Header: 00 00 85 05 00 00 00 48 85 DD>
<U4[3] 9001 9002 9003>
.
```

S5F6 : List Alarm Data (E -> H)

Description:
This message contains the alarm data known to the equipment.

```
2022/2/24 14:36:36.060: Receive S5F6 <Header: 00 00 05 06 00 00 00 2B 11 8B>
<L[3]
 <L[3]
  <B[1] 0x00>
  <U4[1] 9001>
  <A[17] "Pump Press Fressl">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9002>
  <A[22] "Emergency Stop pressed">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9003>
  <A[16] "Heater High Fail">
 >
>
.
```

S5F7 : List Enabled Alarms Request (H -> E)

Description:
List all alarms which are enabled .

```
2022/2/24 14:36:51.378: Send S5F7 <Header: 00 00 85 07 00 00 00 2B 11 8C>
.
```

S5F8 : List Enabled Alarm Data (E -> H)

Description:
This message is similar to S5F6 except that it lists only alarms which are enabled.

```
2022/2/24 14:36:51.384: Receive S5F8 <Header: 00 00 05 08 00 00 00 2B 11 8C>
<L[4]
 <L[3]
  <B[1] 0x00>
  <U4[1] 9001>
  <A[17] "Pump Press Fressl">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9002>
  <A[22] "Emergency Stop pressed">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9003>
  <A[16] "Heater High Fail">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 1000001>
  <A[16] "Heater Low Fail ">
 >
>
.
```

S6F1 : Trace Data Send (E -> H)

Description:
This function sends samples to the host according to the trace setup done by S2F23.
Trace is a time-driven form of equipment status.

```
2022/2/24 16:10:43.192: Receive S6F1 <Header: 00 00 06 01 00 00 00 53 F2 B2>
<L[4]
 <U4[1] 123456789>
 <U4[1] 3>
 <A[16] "2022022416104300">
 <L[9]
  <U1[1] 0>
  <A[0] "">
  <F4[1] 0>
  <U1[1] 0>
  <A[0] "">
  <F4[1] 0>
  <U1[1] 0>
  <A[0] "">
  <F4[1] 0>
 >
>
.
```

S6F2 : Trace Data Acknowledge (H -> E)

Description:
List all alarms which are enabled .

```
2022/2/24 16:10:43.200: Send S6F2 <Header: 00 00 06 02 00 00 00 53 F2 B2>
.
```

S6F11 : Event Report Send (E -> H)

Description:
The purpose of this message is for the equipment to send a defined, linked, and enabled group of reports to the host upon the occurrence of an event(CEID).

```
2022/2/24 16:12:46.967: Receive S6F11 <Header: 00 00 86 0B 00 00 00 53 F2 BB>
<L[3]
 <U2[1] 1>
 <U4[1] 101>
 <L[1]
  <L[2]
   <U4[1] 7001>
   <L[2]
    <U1[1] 0>
    <A[0] "">
   >
  >
 >
>
.
```

S6F12 : Event Report Acknowledge (H -> E)

Description:
Acknowledge or error for Event Report Send(S6F11).

```
2022/2/24 16:12:46.971: Send S6F12 <Header: 00 00 06 0C 00 00 00 53 F2 BB>
<B[1] 0x00>
.
```

S6F15 : Event Report Request (H -> E)

Description:
The purpose of this message is for the host to demand a given report group form the equipment.

```
2022/2/24 14:41:39.817: Send S6F15 <Header: 00 00 86 0F 00 00 00 2B 11 91>
<U4[1] 103>
.
```

S6F16 : Event Report Data (E -> H)

Description:
EQ sends reports linked to given CEID to host.

```
2022/2/24 14:41:39.817: Receive S6F16 <Header: 00 00 06 10 00 00 00 2B 11 91>
<L[3]
 <U2[1] 0>
 <U4[1] 103>
 <L[1]
  <L[2]
   <U4[1] 7004>
   <L[11]
    <F4[1] 0>
    <U1[1] 0>
    <U2[1] 0>
    <U4[1] 0>
    <F4[1] 0>
    <I1[1] 0>
    <I2[1] 0>
    <I4[1] 0>
    <A[0] "">
    <Boolean[1] 0x0>
    <U1[1] 0>
   >
  >
 >
>
.
```

S6F19 : Individual Report Request (H -> E)

Description:
Host requests a Report for the specified RPTID.

```
2022/2/24 14:44:55.416: Send S6F19 <Header: 00 00 86 13 00 00 00 2B 11 93>
<U4[1] 7004>
.
```

S6F20 : Individual Report Data Event Report Data (E -> H)

Description:
EQ sends variable data defined for the given to the host.

```
2022/2/24 14:44:55.416: Receive S6F20 <Header: 00 00 06 14 00 00 00 2B 11 93>
<L[11]
 <F4[1] 0>
 <U1[1] 0>
 <U2[1] 0>
 <U4[1] 0>
 <F4[1] 0>
 <I1[1] 0>
 <I2[1] 0>
 <I4[1] 0>
 <A[0] "">
 <Boolean[1] 0x0>
 <U1[1] 0>
>
.
```

S7F1 : Process Program Load Inquire (H -> E)

Description:
The sender requests permission to send a process program.

```
2022/2/24 14:51:29.744: Send S7F1 <Header: 00 00 87 01 00 00 00 2B 11 96>
<L[2]
 <A[3] "PP1">
 <U4[1] 21>
>
.
```

S7F2 : Process Program Load Grant (E -> H)

Description:
The receiver grants permission to send a process program.

```
2022/2/24 14:51:29.744: Receive S7F2 <Header: 00 00 07 02 00 00 00 2B 11 96>
<B[1] 0x00>
.
```

PPGNT	Description
0x00	Normal. Permission granted.
0x01	Already have
0x02	No space
0x03	Invalid PPID
0x04	Busy.
0x05	Will not accept
0x06	Other error

S7F5 : Process Program Request (H -> E)

Description:
Request the other end of the link to send a specified process program from its library.

```
2022/2/24 14:53:38.416: Send S7F5 <Header: 00 00 87 05 00 00 00 2B 11 99>
<A[3] "PP1">
.
```

S7F6 : Process Program Data (E -> H)

Description:
Send the requested process program.

```
2022/2/24 14:53:38.417: Receive S7F6 <Header: 00 00 07 06 00 00 00 2B 11 99>
<L[2]
 <A[3] "PP1">
 <A[21] "This is a test recipe">
>
.
```

S7F19 : Process Program Dir. Request (H -> E)

Description:
The host requests a directory of the process program currently in the EQ's library.

```
2022/2/24 14:55:09.647: Send S7F19 <Header: 00 00 87 13 00 00 00 2B 11 9A>
.
```

S7F20 : Current Process Program Data (E -> H)

Description:
The EQ sends a directory of its process program library. If the library is empty, the list will be zero length.

```
2022/2/24 14:55:09.649: Receive S7F20 <Header: 00 00 07 14 00 00 00 2B 11 9A>
<L[12]
 <A[4] "1234">
 <A[2] "20">
 <A[3] "222">
 <A[2] "99">
 <A[3] "PP1">
 <A[13] "PP1_Formatted">
 <A[3] "PP2">
 <A[7] "PP2.pjb">
 <A[13] "PP2_Formatted">
 <A[47] "PP3_1234567890_1234567890_1234567890_1234567890">
 <A[10] "PP3_BINARY">
 <A[13] "PP4_Formatted">
>
.
```

S7F25 : Formatted Process Program Request (H -> E)

Description:

The host requests a process program currently in the EQ's library.

```
2022/2/24 14:56:37.776: Send S7F25 <Header: 00 00 87 19 00 00 00 2B 11 A1>
<A[13] "PP1_Formatted">
.
```

S7F26 : Formatted Process Program Data (E -> H)

Description:

The EQ sends a formatted process program data to the host.

```
2022/2/24 14:56:37.778: Receive S7F26 <Header: 00 00 07 1A 00 00 00 2B 11 A1>
<L[4]
 <A[13] "PP1_Formatted">
 <A[13] "SMT Equipment">
 <A[7] "1.0.0.0">
 <L[1]
  <L[2]
   <U2[1] 2>
   <L[9]
    <U1[1] 0>
    <U2[1] 1>
    <U4[1] 1>
    <I1[1] 100>
    <I2[1] 1000>
    <I4[1] 1000000>
    <F4[1] 0.1>
    <Boolean[1] 0x0>
    <A[50] "A123456789_123456789_123456789_123456789_123456789">
   >
  >
 >
>
.
```

S9F7 : Invalid Data

Description:
This message indicates that the stream and function were recognized, but the associated data format could not be interpreted.

```
2022/2/24 16:20:14.221: Receive S9F7 <Header: 00 00 09 07 00 00 00 09 78 97>
<B[10] 0x00 0x00 0x86 0x01 0x00 0x00 0x00 0x7E 0xF9 0xEE>
.
```

S9F9 : Transaction Timer Timeout

Description:
This message indicates that a transaction (receive) timer has timed out and that the corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.

```
16:21:03.257: T3 Timeout(8321520).
2022/2/24 16:21:03.258: Send S9F9 <Header: 00 00 09 09 00 00 00 7E F9 F1>
<B[10] 0x00 0x00 0x09 0x09 0x00 0x00 0x00 0x7E 0xF9 0xEE>
.
```

S10F1 : Terminal Request

Description:
A terminal text message to the host.

```
2022/2/24 16:26:53.637: Receive S10F1 <Header: 00 00 8A 01 00 00 00 09 78 9F>
<L[2]
 <B[1] 0x00>
 <A[26] "Hi ! This is Eqp's Message">
>
.
```

S10F2 : Terminal Request Acknowledge

Description:
Acknowledge or error for terminal request (S10F1).

```
2022/2/24 16:26:53.639: Send S10F2 <Header: 00 00 0A 02 00 00 00 09 78 9F>
<B[1] 0x00>
.
```

S10F3 : Terminal Display, Single (H -> E)

Description:
Data to be displayed.

```
2022/2/24 15:33:09.688: Send S10F3 <Header: 00 00 8A 03 00 00 00 2B 11 A2>
<L[2]
 <B[1] 0x00>
 <A[16] "TerminalDataTest">
>
.
```

TID	Description
Any	Equipment displays the text on its CRT.

Message	Description
Any	Equipment displays the text
0XX	File MappingTable in folder ini or EQ Set

S10F4 : Terminal Request Acknowledge (E -> H)

Description:
Acknowledge or error for terminal request (S10F3).

```
2022/2/24 15:33:09.689: Receive S10F4 <Header: 00 00 0A 04 00 00 00 2B 11 A2>
<B[1] 0x00>
.
```

ACKC10	Description
0x00	Normal.

FVI SECS/GEM INFORMATION

ReelID Send LOG

Description:

Equipment will send ReelID (4001) attached to the ReelID_Send (101) event to host, host need to send back APN, DATECODE and LOT6 to equipment.

```
2022/5/26 11:34:53.773: Receive S6F11 <Header: 00 00 86 0B 00 00 00 01 8F 63>
<L[3]
 <U2[1] 5>
 <U4[1] 101>
 <L[1]
  <L[2]
   <U4[1] 101>
   <L[1]
    <A[14] "TR00L2X2P9L101">
   >
  >
 >
>
.
2022/5/26 11:34:53.777: Send S6F12 <Header: 00 00 06 0C 00 00 00 01 8F 63>
<B[1] 0x00>
.
```

```
2022/5/26 11:34:58.941: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[20] "REELID_VERIFY_RESULT">
 <L[3]
  <L[2]
   <A[3] "APN">
   <A[9] "339S00797">
  >
  <L[2]
   <A[8] "DATECODE">
   <A[4] "2221">
  >
  <L[2]
   <A[4] "LOT6">
   <A[6] "L2X2P9">
  >
 >
>
.
2022/5/26 11:34:58.948: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
```

Recipe Change LOG

Description:

Host order Recipe Change command to equipment, equipment will send CurrentRecipeName(4005) attached to the Recipe_Change(102) event to Host when recipe change finished.

```
2022/5/26 11:42:31.788: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[13] "RECIPE_CHANGE">
 <L[1]
  <L[2]
   <A[11] "RECIPE_NAME">
   <A[8] "Recipe_A">
  >
 >
>
.
2022/5/26 11:42:31.789: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

```
2022/5/26 11:42:35.189: Receive S6F11 <Header: 00 00 86 0B 00 00 00 01 8F 6B>
<L[3]
 <U2[1] 6>
 <U4[1] 102>
 <L[1]
  <L[2]
   <U4[1] 102>
   <L[1]
    <A[8] "Recipe_A">
   >
  >
 >
>
.
2022/5/26 11:42:35.192: Send S6F12 <Header: 00 00 06 0C 00 00 00 01 8F 6B>
<B[1] 0x00>
.
```

Equipment Remote Control LOG

Description:
Host order REMOTE-START, REMOTE-STOP or REMOTE-RESET command to equipment, equipment will send Equipment_RemoteControlStatus(1001) and Equipment_RemoteControlResult (1002) attached to the Equipment_RemoteControl(103) event to Host.

REMOTE START LOG

```
2022/5/26 13:48:10.911: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[24] "EQUIPMENT_REMOTE_CONTROL">
 <L[1]
  <L[2]
   <A[30] "EQUIPMENT_REMOTECONTROL_STATUS">
   <A[12] "REMOTE-START">
  >
 >
>
.
2022/5/26 13:48:10.917: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

```
2022/5/26 13:48:14.130: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 2A>
<L[3]
 <U2[1] 1>
 <U4[1] 103>
 <L[1]
  <L[2]
   <U4[1] 103>
   <L[2]
    <A[12] "REMOTE-START">
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/5/26 13:48:14.132: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 2A>
<B[1] 0x00>
.
```

※ Equipment_RemoteControlResult will show SUCCESS or FAIL

REMOTE STOP LOG

```
2022/5/26 13:50:19.399: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[24] "EQUIPMENT_REMOTE_CONTROL">
 <L[1]
  <L[2]
   <A[30] "EQUIPMENT_REMOTECONTROL_STATUS">
   <A[11] "REMOTE-STOP">
  >
 >
>
.
2022/5/26 13:50:19.400: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

```
2022/5/26 13:50:22.807: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 2D>
<L[3]
 <U2[1] 2>
 <U4[1] 103>
 <L[1]
  <L[2]
   <U4[1] 103>
   <L[2]
    <A[11] "REMOTE-STOP">
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/5/26 13:50:22.811: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 2D>
<B[1] 0x00>
.
```

※ Equipment_RemoteControlResult will show SUCCESS or FAIL

REMOTE RESET LOG

```
2022/5/26 13:52:19.223: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[24] "EQUIPMENT_REMOTE_CONTROL">
 <L[1]
  <L[2]
   <A[30] "EQUIPMENT_REMOTECONTROL_STATUS">
   <A[12] "REMOTE-RESET">
  >
 >
>
.
2022/5/26 13:52:19.224: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

```
2022/5/26 13:52:29.784: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 2F>
<L[3]
 <U2[1] 3>
 <U4[1] 103>
 <L[1]
  <L[2]
   <U4[1] 103>
   <L[2]
    <A[12] "REMOTE-RESET">
    <A[4] "FAIL">
   >
  >
 >
>
.
2022/5/26 13:52:29.786: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 2F>
<B[1] 0x00>
.
```

※ Equipment_RemoteControlResult will show SUCCESS or FAIL

Equipment Local Control LOG

Description:
Host order LOCAL-START, LOCAL -STOP or LOCAL -RESET command to equipment, equipment will send Equipment_LocalControlStatus(1003) and Equipment_LocalControlResult (1004) attached to the Equipment_LocalControl(104) event to Host.

LOCAL START LOG

```
2022/5/26 13:55:04.815: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 32>
<L[3]
 <U2[1] 4>
 <U4[1] 104>
 <L[1]
  <L[2]
   <U4[1] 104>
   <L[2]
    <A[11] "LOCAL-START">
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/5/26 13:55:04.818: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 32>
<B[1] 0x00>
.
```

※ Equipment_LocalControlResult will show SUCCESS or FAIL

LOCAL STOP LOG

```
2022/5/26 13:56:01.624: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 33>
<L[3]
 <U2[1] 5>
 <U4[1] 104>
 <L[1]
  <L[2]
   <U4[1] 104>
   <L[2]
    <A[10] "LOCAL-STOP">
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/5/26 13:56:01.626: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 33>
<B[1] 0x00>
.
```

※ Equipment_LocalControlResult will show SUCCESS or FAIL

LOCAL RESET LOG

```
2022/5/26 13:56:33.471: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 34>
<L[3]
 <U2[1] 6>
 <U4[1] 104>
 <L[1]
  <L[2]
   <U4[1] 104>
   <L[2]
    <A[11] "LOCAL-RESET">
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/5/26 13:56:33.474: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 34>
<B[1] 0x00>
```

※ Equipment_RemoteControlResult will show SUCCESS or FAIL

End Lot LOG

Description:

Equipment will send Reel_ID(4001) and Actual_DieCount(4007) attached to the EndLot(105) event to host, then host need send back expected DieCount and compare result to equipment.

```
.
2022/8/10 15:47:03.137: Receive S6F11 <Header: 00 00 86 0B 00 00 00 77 CD E0>
<L[3]
 <U4[1] 1>
 <U4[1] 105>
 <L[1]
  <L[2]
   <U4[1] 105>
   <L[2]
    <A[14] "TR00L2X2P9L101">
    <A[3] "450">
   >
  >
 >
>
.
2022/8/10 15:47:03.139: Send S6F12 <Header: 00 00 06 0C 00 00 00 77 CD E0>
<B[1] 0x00>
.
```

```
2022/5/26 13:58:21.790: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[23] "DIECOUNT_COMPARE_RESULT">
 <L[2]
  <L[2]
   <A[17] "EXPECTED_DIECOUNT">
   <A[3] "450">
  >
  <L[2]
   <A[23] "DIECOUNT_COMPARE_RESULT">
   <A[4] "PASS">
  >
 >
>
.
2022/5/26 13:58:21.791: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※ DIECOUNT COMPARE RESULT please give PASS or NG to it.

Reel Current Status LOG

Description:

Equipment will send Reel_Status(1005) attached to the Reel_currentStatus(106) event to host when Reel has been add or remove.

ADD LOG

```
2022/5/26 14:14:54.855: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 47>
<L[3]
 <U2[1] 8>
 <U4[1] 106>
 <L[1]
  <L[2]
   <U4[1] 106>
   <L[1]
    <A[22] "The Reel has been add.">
   >
  >
 >
>
.
2022/5/26 14:14:54.859: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 47>
<B[1] 0x00>
.
```

REMOVE LOG

```
2022/5/26 14:15:06.165: Receive S6F11 <Header: 00 00 86 0B 00 00 00 13 D0 48>
<L[3]
 <U2[1] 9>
 <U4[1] 106>
 <L[1]
  <L[2]
   <U4[1] 106>
   <L[1]
    <A[25] "The Reel has been remove.">
   >
  >
 >
>
.
2022/5/26 14:15:06.169: Send S6F12 <Header: 00 00 06 0C 00 00 00 13 D0 48>
<B[1] 0x00>
.
```

Go remote(local) switch LOG

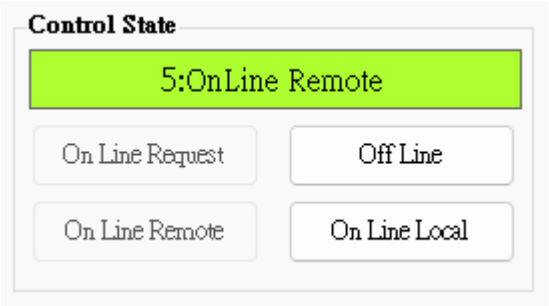
Description:

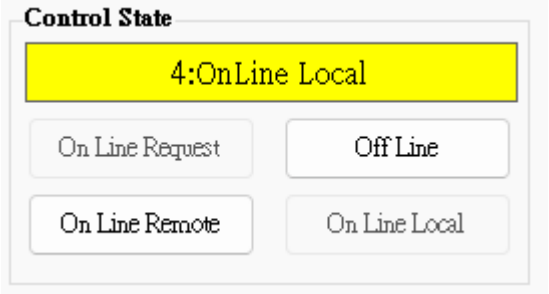
Host sned GO-REMOTE(GO-LOCAL) command to equipment, equipment will change control state.

#When equipment in local state, it will only accept “GO-REMOTE” command and block other S2F41 command.

GO REMOTE LOG

```
.
2022/8/10 14:25:40.846: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[9] "GO-REMOTE">
 <L[0]
 >
>
.
2022/8/10 14:25:40.852: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```



GO LOCAL LOG

```
.
2022/8/10 14:27:18.342: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[8] "GO-LOCAL">
 <L[0]
 >
>
.
2022/8/10 14:27:18.344: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x02>
 <L[0]
 >
>
.
```